CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 10



By- Vishvadeep Gothi sir

Recap of Previous Lecture



Topic — Array Access with Cache

Topic — Multilevel Cache

Topics to be Covered



Topic Multilevel Cache

Topic Cache Inclusion Policy

Topic Magnetic Disk

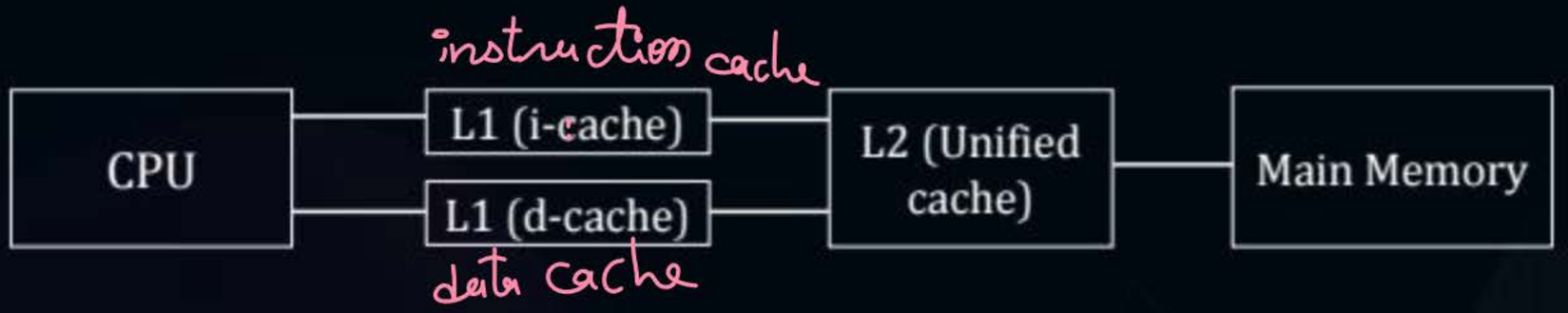


8:15 PM – 10:15 PM

COA



Topic : Dual Cache



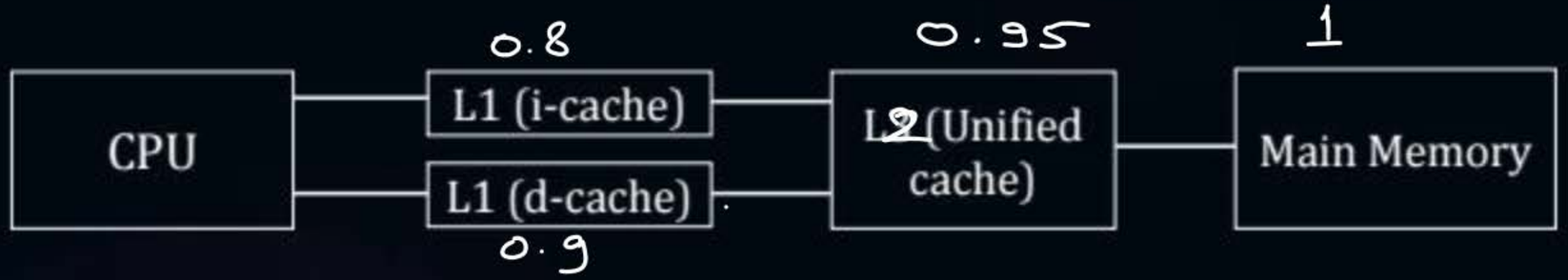
$$t_{avg\ inst^n} = H_{1i} * t_{1i} + (1 - H_{1i}) \left[H_2 * (t_{1i} + t_2) + (1 - H_2)(t_{1i} + t_2 + t_{mm}) \right]$$

$$t_{avg\ data} = H_{1d} * t_{1d} + (1 - H_{1d}) \left[H_2 * (t_{1d} + t_2) + (1 - H_2)(t_{1d} + t_2 + t_{mm}) \right]$$

$$t_{avg} = \text{fraction of } \underset{\text{access}}{\text{inst}^n} * t_{avg\ inst^n} + \text{fraction of data access} * t_{avg\ data}$$

[NAT]

#Q. The multilevel memory hierarchy is given.



The hit ratio of L1i, L1d, L2 and main memory are 0.8, 0.9, 0.95 and 1.0 respectively. The access times of respective memories are 10ns, 10ns, 50ns and 500ns. Among total memory references 60% of them are for data.

1. Average memory access time for only instructions access 25 ns
2. Average memory access time for only data access 17.5 ns
3. Average memory access time 20.5 ns

$$t_{avg\ inst^n} = 10 + 0.2\left[50 + 0.05 * 500\right] = 25\ ns$$

$$t_{avg\ data} = 10 + 0.1\left[50 + 0.05 * 500\right] = 17.5\ ns$$

$$t_{avg} = 0.4 * 25 + 0.6 * 17.5\ ns$$

$$= 20.5\ ns$$

[NAT]

GATE – 2017

#Q. The read access times and the hit ratios for different caches in a memory hierarchy are as given below:

Cache	Read access time (in nanoseconds)	Hit Ratio
I-cache	2	0.8
D-cache	2	0.9
L2-cache	8	0.9

The read access time of main memory in 90nanoseconds. Assume that the caches use the referred-word-first read policy and the write-back policy. Assume that all the caches are direct mapped caches. Assume that the dirty bit is always 0 for all the blocks in the caches. In execution of a program, 60% of memory reads are for instruction fetch and 40% are for memory operand fetch. The average read access time in nanoseconds (up to 2 decimal places) is 4.72 ?

$$T_{avg\ inst^n} = 2 + 0.2\left[8 + 0.1 * 90\right] = 5.4\ ns$$

$$T_{avg\ data} = 2 + 0.1\left[8 + 0.1 * 90\right] = 3.7\ ns$$

$$T_{avg} = 0.6 * 5.4 * 0.4 * 3.7$$

$$= 4.72\ ns$$

[NAT]

extra time

#Q. Consider a program execution which has 36% instructions for load and store. The CPI without memory stalls is 2. The program experiences 2% miss for instruction cache and 4% miss for data cache. The cache miss penalty is 200 cycles.

1. CPI with memory stalls 8.88 cycles
2. Performance gain (speed up) of perfect cache as compared to cache with stalls ⇒ 4.44

stalls due to instn access = 1 * 0.02 * 200 = 4 cycles

stalls due to data access = 0.36 * 0.04 * 200 = 2.88 cycles

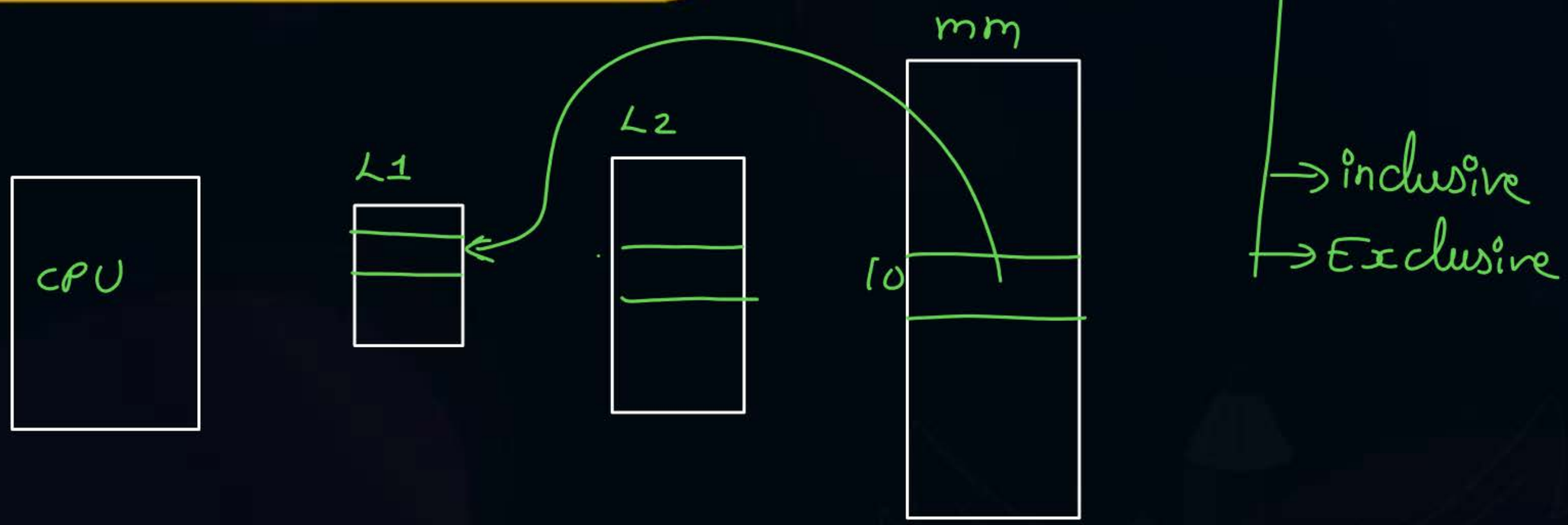
CPI = 2 + 4 + 2.88 = 8.88 cycles

2. performance gain $= \frac{8.88}{2}$

$= 4.44$



Topic : Cache Inclusion Policy





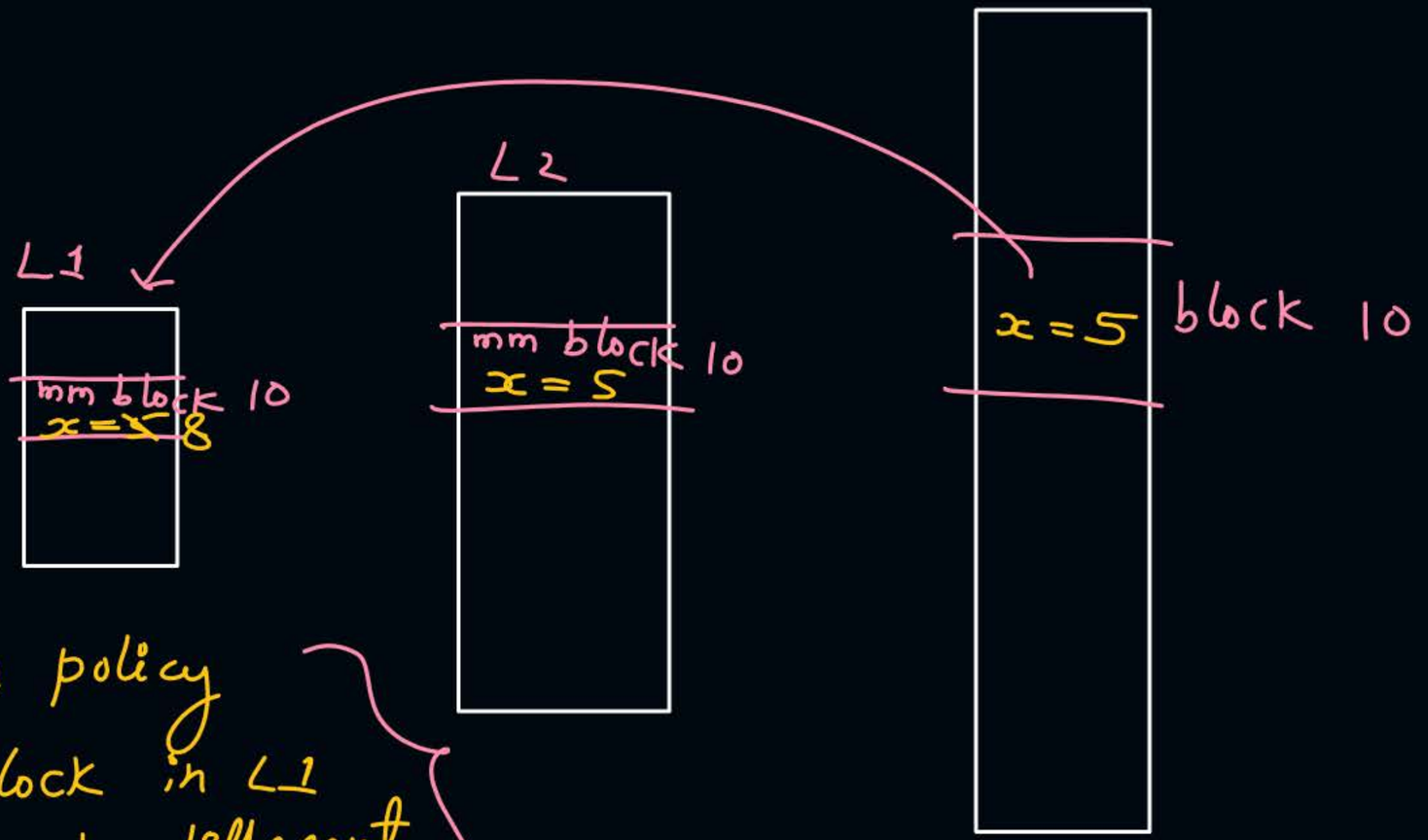
Topic : ~~Inclusion~~ Inclusive Policy

all mm blocks which are present in L1, must be present in L2 too.

(L2 is inclusive of L1)

$$(\text{Size of } L2) \geq (\text{Size of } L1)$$

$$(\text{Content of } L1) \cap (\text{Content of } L2) \Rightarrow L1$$



In inclusion policy value in a block in L1 and L2 can be different.

→ values in blocks in L1 and L2 also must be same

⇓

Value inclusion policy



Topic : Inclusion Policy

CPU wants to read a value x from mem.

1. Hit in L1 ⇒ CPU reads x from L1.

2. Miss in L1 & Hit in L2 ⇒ CPU reads x from L2.
   missed block is copied from L2 to L1.
   If any block is replaced from L1, then there is no any role of L2.

3. Miss in L1 & Miss in L2 ⇒ CPU reads x from mm
   missed block is copied from mm to L2, then from L2 to L1
   If any block is replaced from L2 then L2 sends a back validation signal to L1 to make this block invalid L1.
   If any block is replaced from L1 then there is no any role of L2.



Topic : Exclusion Policy

all the mm blocks which are present in L1 should not be present in L2.

$$(\text{Content of } L1) \cap (\text{Content of } L2) \Rightarrow \phi$$

L2 is victim cache because it stores all victim blocks of L1.



Topic : Exclusion Policy

CPU wants to read x from mem.

1. Hit in L1 ⇒ CPU reads x from L1

2. Miss in L1 & Hit in L2 ⇒ CPU reads x from L2
   The missed block is moved (removed and copied) from L2 to L1.
   If any block is replaced from L1 then place it into L2.

3. Miss in L1 & Miss in L2 ⇒ CPU reads x from mm
   The missed block is copied from mm to L1.
   If any block is replaced from L1 then place it into L2.

If any block replaced from L2 then no role of L1.

[NAT]

#Q. Consider a direct mapped write back data cache of size 2KB with block size of 128 bytes. The cache is considered to be empty initially. The byte addressable main memory has size 1Mbytes. Further consider that there is an array A[30][25] with each element occupies 4 bytes. The base address of array is $(1A300)_{16}$. The array is accessed 4 times. And between the accesses, there is no any data cache changes happen. Hit ratio of cache for the array access is?

Array size = 30 * 25 = 750 = 750 * 4 = 3000 bytes

no. of blocks to store array = $\left\lceil \frac{3000\,B}{128\,B} \right\rceil$ = 24 blocks

no. of blocks in cache = $\frac{2KB}{128B} = 2^4 = 16$

extra blocks = 24 − 16 = 8

	miss
1st access	24
2nd access	2*8=16
3rd	16
4th	16
Total	72

Total no of mem references = 750 * 4
= 3000

hits = 3000 − 72
= 2928

hit ratio = $\frac{2928}{3000}$

= 0.976
= 97.6 %

[MSQ] [2022 : 2M]

2^{16} B ⇒ add = 16 bits

#Q. Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (i.e., PQRSPQRS...). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

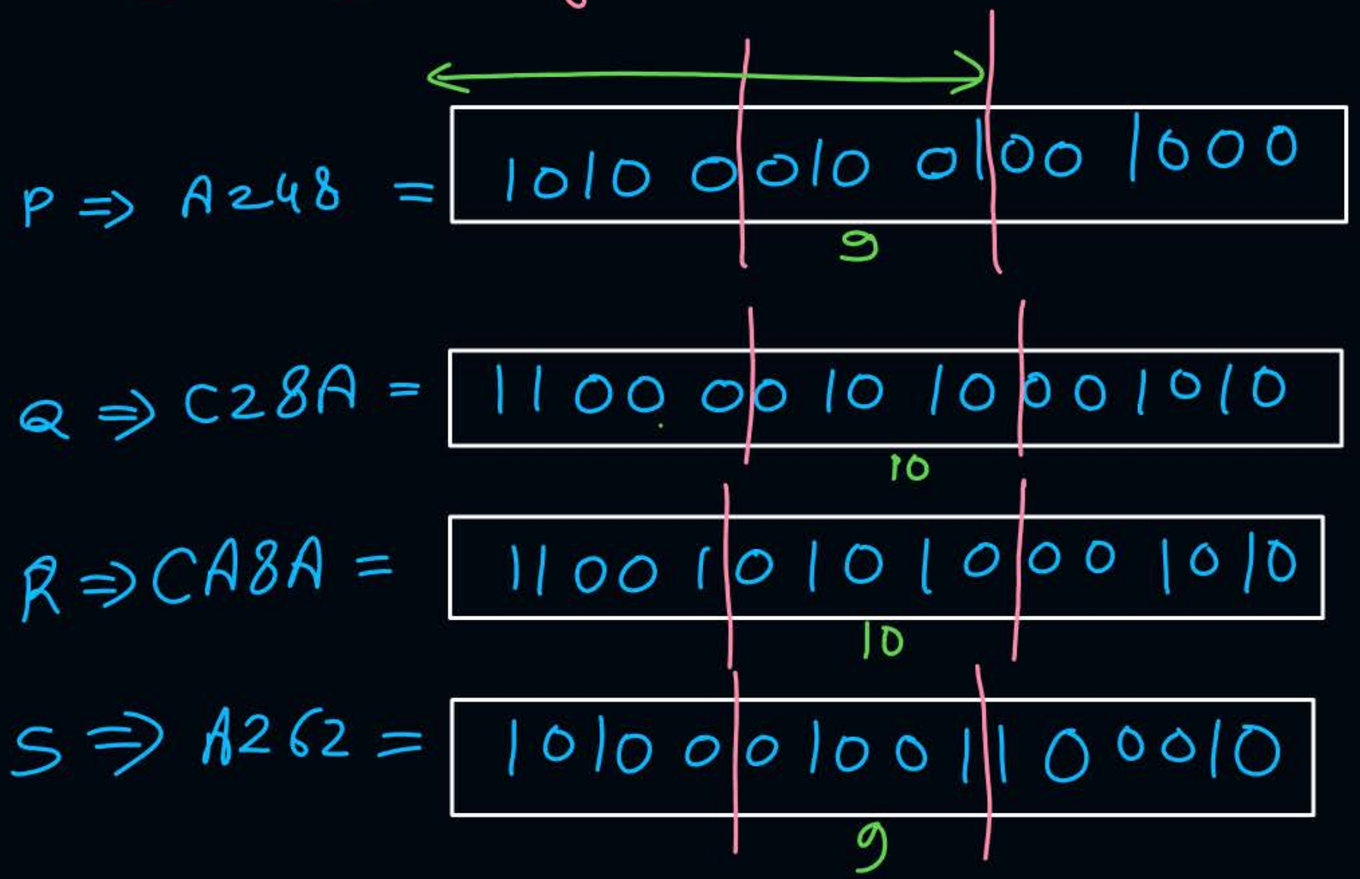
A ✓ Every access to S is a hit.

B ✓ Once P is brought to the cache it is never evicted.

C ✗ At the end of the execution only R and S reside in the cache. → P, R, S will be in mem.

D ✓ Every access to R evicts Q from the cache.





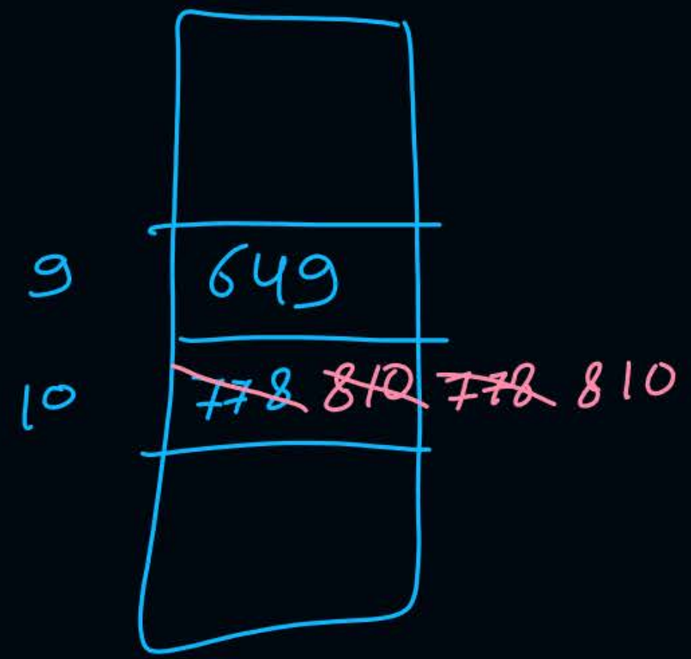
P, S add. are in same mm block

	mm block no.
P ⇒	649
Q ⇒	778
R ⇒	810
S ⇒	649

CPU accesses mm block no.

649, 778, 810, 649

↓ miss ↓ miss ↓ miss ↓ hit





Topic : T_{avg} in Write Back Cache

assume fraction of dirty blocks replaced from cache $\Rightarrow d$

$$T_{avg} = H * t_{cm} + (1-H)\left[t_{cm} + t_{bt} + d * t_{bt}\right]$$

t_{bt} → to bring missed block from mm to cache

$d * t_{bt}$ → write back time



2 mins Summary

Topic Array Access with Cache

Topic Multilevel Cache

Topic Magnetic Disk



Happy Learning

THANK - YOU